

TAI NGUYEN PHU

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EDUCATION

University of Information Technology (UIT - VNUHCM) <i>Bachelor of Computer Science</i>	Sep 2022 - Nov 2025 GPA: 3.5/4.0
Le Khiet High School for the Gifted <i>Information Technology Major</i>	Sep 2019 - Jun 2022 GPA: 3.4/4.0

SKILLS

Languages:	Proficient in Python, C/C++; Basic in JavaScript, React, SQL, Bash, HTML, CSS
BE & Deployment:	REST APIs (FastAPI, Flask); Git, GitHub Actions, Docker, Weights & Biases
AI/ML:	PyTorch, TensorFlow, Pydantic, Pandas, Spark, Scikit-learn, NumPy, Chroma, FAISS
NLP/CV:	Transformers, PEFT, SFT, SentencePiece, LangChain, LangGraph, OpenCV, YOLO, Detectron2
Courseworks:	Linear Algebra, Calculus, Probability & Statistics, Data Structures & Algorithms

PROJECTS

Multimodal Agentic RAG System for PDF documents 🔗	Jun 2025 - Aug 2025
- Designed and implemented an end-to-end Agentic RAG workflow with LangGraph, building modular nodes for generation, retrieval, analysis, rewriting, and synthesis. - Engineered a multimodal PDF ingestion pipeline leveraging OpenAI Vision and Docling to enable structured extraction and intelligent querying across text and images. - Integrated Milvus vector store with BM25 hybrid retrieval and BGE-M3 embeddings, orchestrated and monitored via Docker and Attu for scalable deployment. - Enhanced transparency and multi-turn interaction by implementing conversational memory and detailed agent step tracing within the workflow. - Delivered a Flask-based web application supporting PDF/image uploads, model selection, real-time workflow visualization, and an interactive chat interface.	
Vietnamese Legal Document Semantic Retrieval System 🔗	Mar 2025 - Apr 2025
- Fine-tuned a multilingual SBERT model on a corpus of 100K+ Vietnamese legal documents using MNR and advanced batch sampling strategies, enhancing semantic retrieval performance for domain-specific queries. - Attained strong benchmark results with NDCG@10 of 60.4% and MAP@10 of 53.6% on MTEB (BKAI Legal Retrieval), demonstrating effectiveness in legal information retrieval tasks. - Built a complete pipeline encompassing data preprocessing, contrastive fine-tuning, task-specific MTEB evaluation, and deployment workflows. - Engineered a FAISS-based retrieval system with an interactive Gradio interface, enabling efficient semantic search and user accessibility. - Containerized the entire system with Docker to ensure reproducibility, portability, and seamless deployment across environments.	
Glassdoor 2024 Data Science Job Salary Prediction 🔗	Dec 2024 - Jan 2025
- Crawled 2024 Glassdoor Data Science jobs and performed comprehensive preprocessing, EDA, and feature engineering. - Achieved $R^2 = 0.82$ using XGBoost, significantly outperforming the Linear Regression baseline ($R^2 = 0.71$). - Enhanced XGBoost predictive performance and robustness through advanced hyperparameter tuning with Optuna. - Experimented with LLM-based regression approaches via few-shot prompting with OpenAI's GPT-5 SDK and supervised fine-tuning (SFT + QLoRA) with LLaMA 3.1. - Benchmarked model performance across traditional ML, few-shot LLMs, and fine-tuned LLMs; built a salary prediction Streamlit app based on the three modeling approaches.	

CERTIFICATIONS

Natural Language Processing Specialization 🔗
Deep Learning Specialization 🔗
Machine Learning Specialization 🔗
IELTS Academic | 7.0 *Overall* 🔗

ACHIEVEMENTS

Kaggle Competition: "Credit Risk Model Stability" | *Rank* 1096/3858 🔗
UIT Global Scholarship 🔗